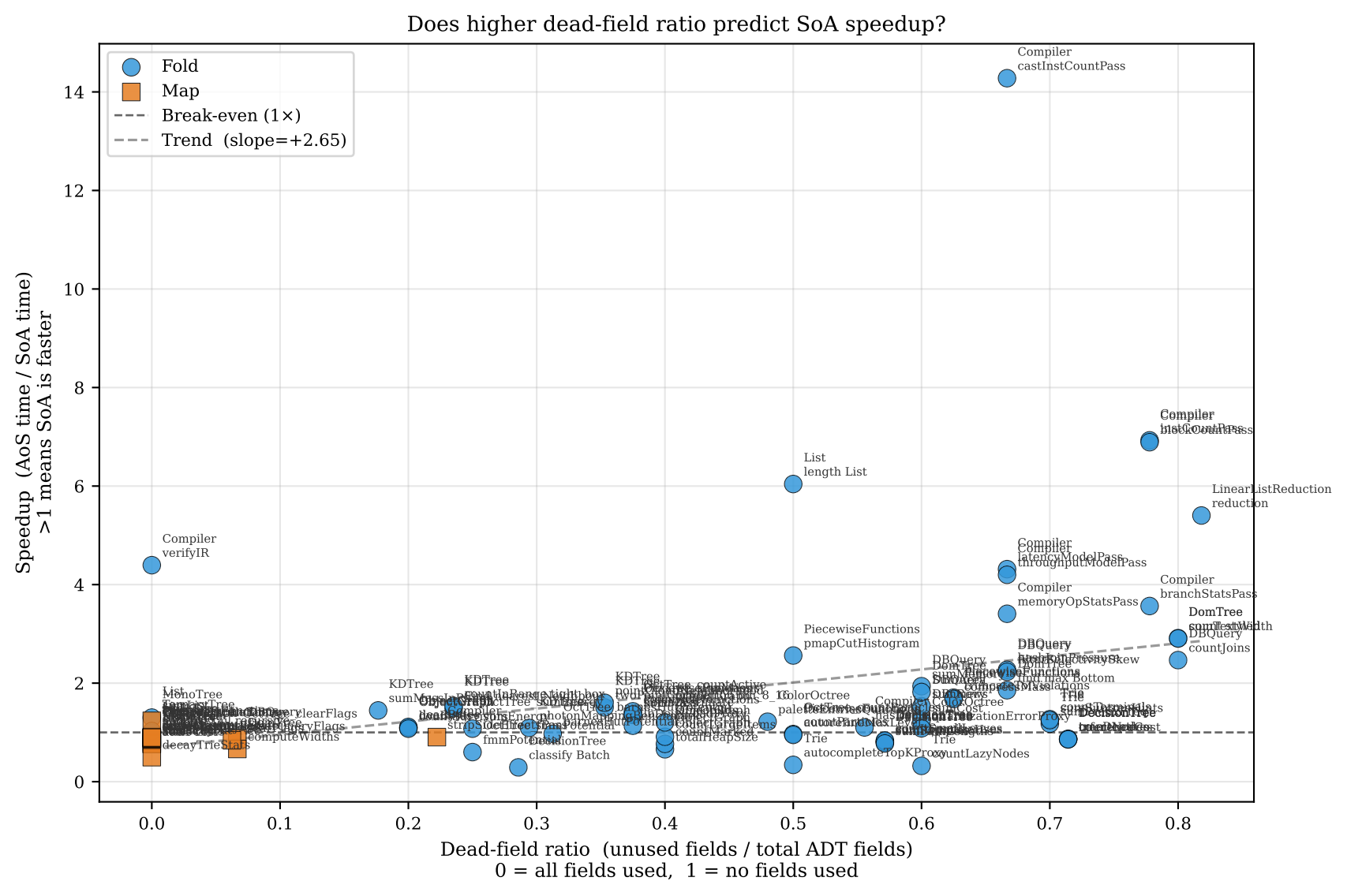

Does higher dead-field ratio predict SoA speedup?
Fold
Map
Break-even (1×)
Trend (slope=+2.65)
Speedup (AoS time / SoA time)
>1 means SoA is faster
Dead-field ratio (unused fields / total ADT fields)
0 = all fields used, 1 = no fields used
0
2
4
6
8
10
12
14
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Compiler
castInstCountPass
List
length List
LinearListReduction
reduction
Compiler
verifyIR
Compiler
latencyModelPass
throughputModelPass
Compiler
memoryOpStatsPass
Compiler
branchStatsPass
PiecewiseFunctions
pmapCutHistogram
DBQuery
countJoins
KDTree
sumNodeCount
ColorOctree
DecisionTree
classify Batch
Trie
autocomplete
TopKProxy
countLazyNodes
decayTrieStats
computeWidths